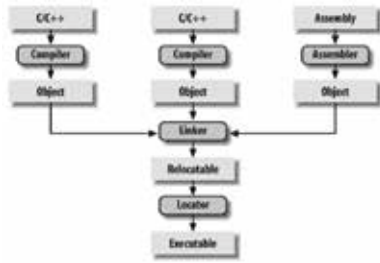


In this section we'll talk about the compilers, linkers and assemblers along with build systems which will handle all these tasks transparently for you with the only requirement being the configuration. Downloading the executable to the target system is usually carried out through either USB, serial IO or network. The host requires the target boards interface drivers to be present so that it can send data to the board. The interface is then responsible for moving the code and data to the correct place in the memory.



A generic embedded software development process

Cross Compilers

Microcontrollers do not have a full-fledged operating systems which can run compilers for the code for that particular platform. This is where cross compilers come in, a cross compiler is capable of creating executable code for a target platform which is different from the platform on which it is running. Eg. you're using your x86-64 based system to generate code for an ARM processor.

There are a multitude of cross compilers available in the market today. One of them is GCC, a collection of compilers from GNU. GCC can be setup to cross compile to a different platform. GCC needs a compiled executable of binutils to be made available for each of the target platform. Another important part is the GNU Assembler. binutils must be compiled with the `--target` switch set to the desired target platform in the configure script. GCC also has to be configured with the same target option. Cross compiling GCC requires that at least a subset of the target's C standard library be available on the host. You can choose to have the full C library but a better alternative is to use newlib, which is a smaller, light weight C library containing only the important modules required to compile the C source code that you provide.

Another such compiler is ELLCC which uses Clang and LLVM compiler infrastructure to provide an easy to use multi-target cross compilation environment. It requires ecc, the ELLCC C/C++ compiler, binutils, libecc and QEMU for working. GNU Gnat is a compiler for Ada, it is under development and has an engaging community.